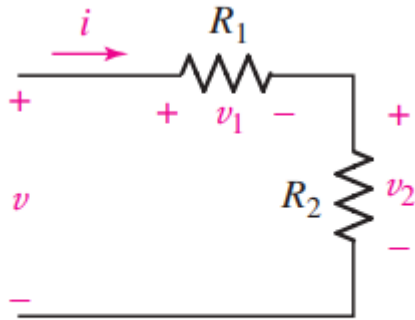
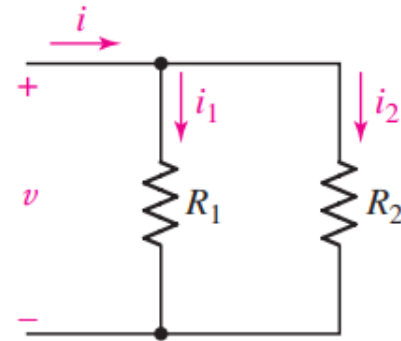


VOLTAGE AND CURRENT DIVISION

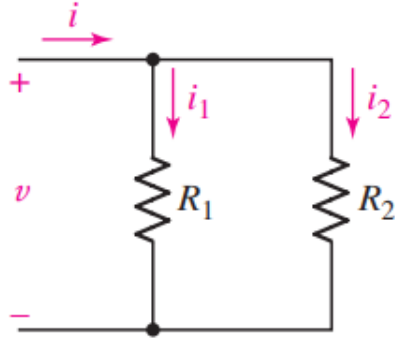


■ **FIGURE 3.34** An illustration of voltage division.



■ **FIGURE 3.37** An illustration of current division.

VOLTAGE AND CURRENT DIVISION



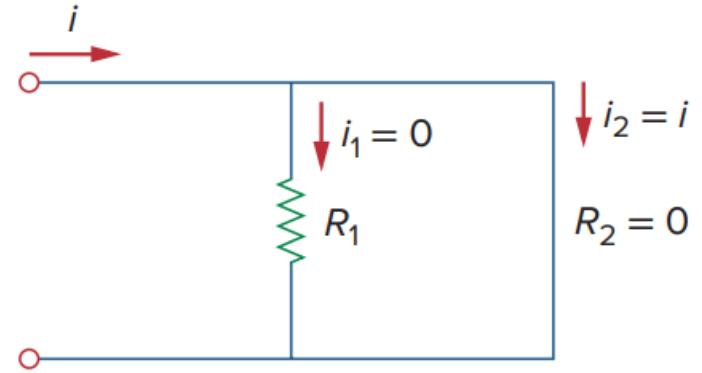
■ **FIGURE 3.37** An illustration of current division.

$$i_k = i \frac{\frac{1}{R_k}}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}}$$

Short Circuit

$$I_1 = \frac{R_2}{R_1 + R_2} I = \frac{0}{R_1 + 0} I = 0$$

$$I = I_1 + I_2 = 0 + I_2$$

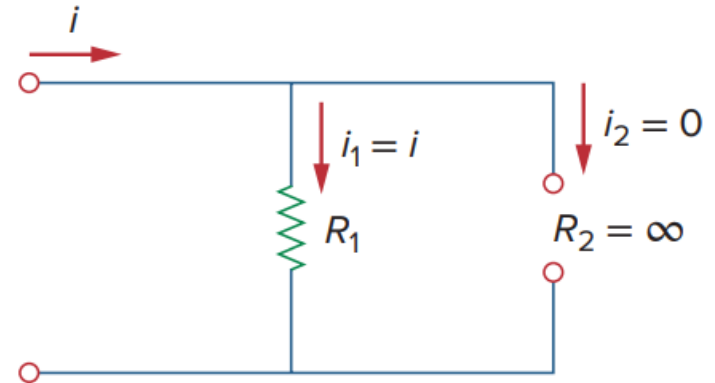


(a)

Open Circuit

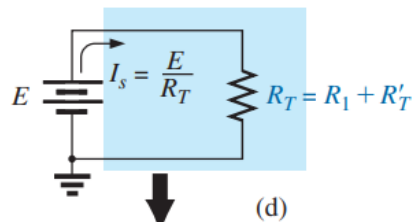
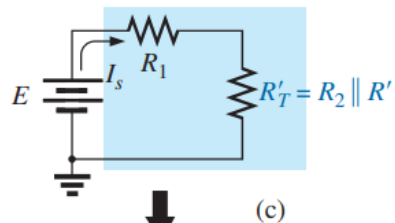
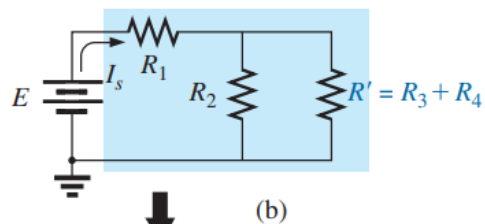
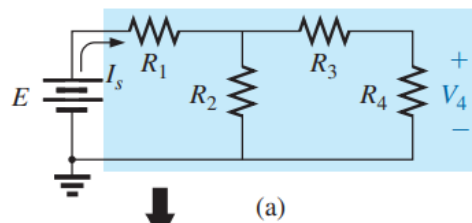
$$I_2 = \frac{R_1}{R_1 + R_2} I = \frac{R_1}{R_1 + \infty} I = 0$$

$$I = I_1 + I_2 = I_1 + 0$$

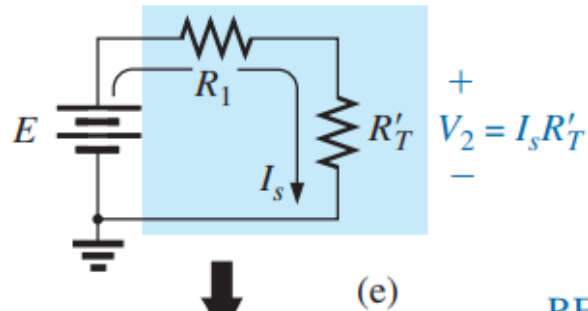


(b)

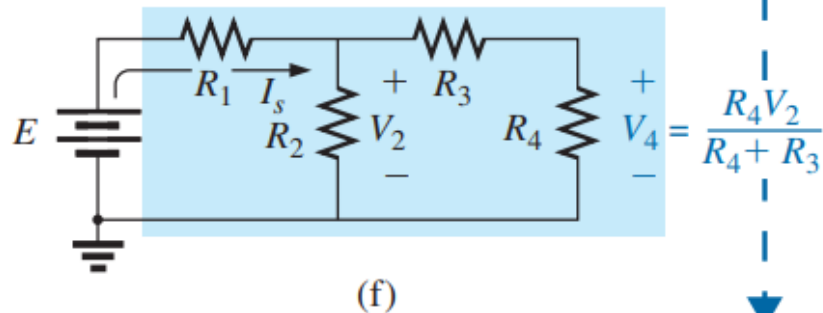
REDUCE AND RETURN APPROACH



REDUCE

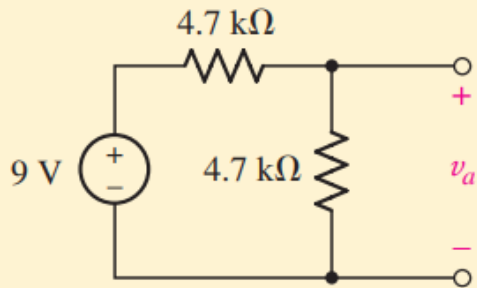


RETURN



$$V_4 = \frac{R_4 V_2}{R_4 + R_3}$$

voltages are defined across two clearly marked terminals, not any particular point.



■ **FIGURE 3.40** A simple circuit with a voltage v_a defined between two terminals.

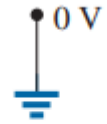
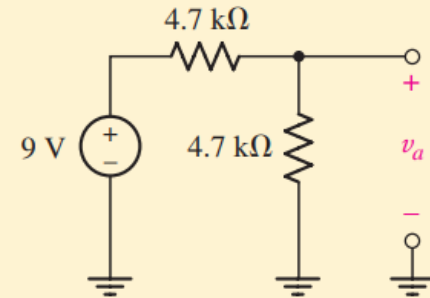
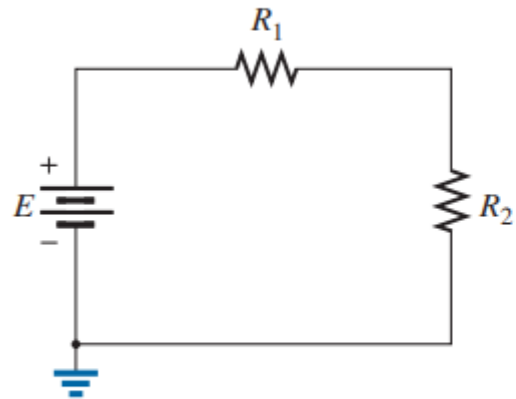


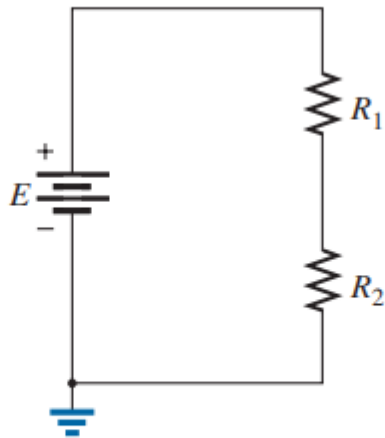
FIG. 5.47
Ground potential.



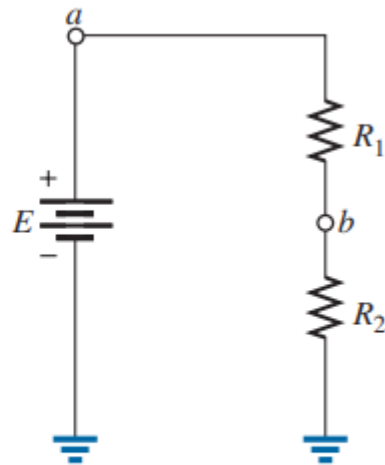
■ **FIGURE 3.42** The circuit of Fig. 3.40, redrawn using the earth ground symbol. The rightmost ground symbol is redundant; it is only necessary to label the positive terminal of v_a ; the negative reference is then implicitly ground, or zero volts.



(a)



(b)



(c)

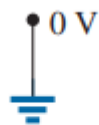
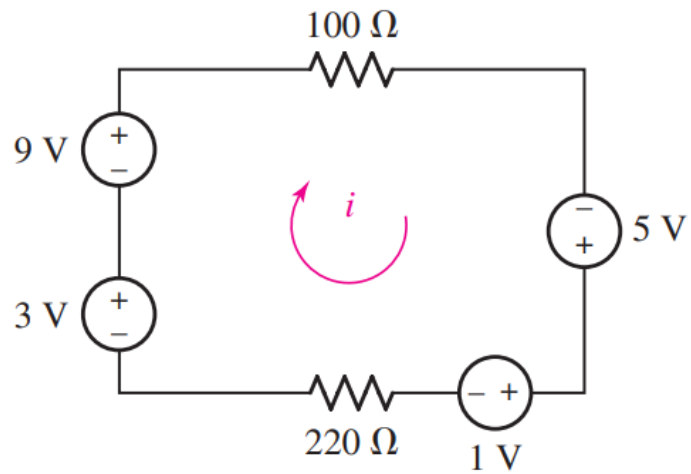


FIG. 5.47

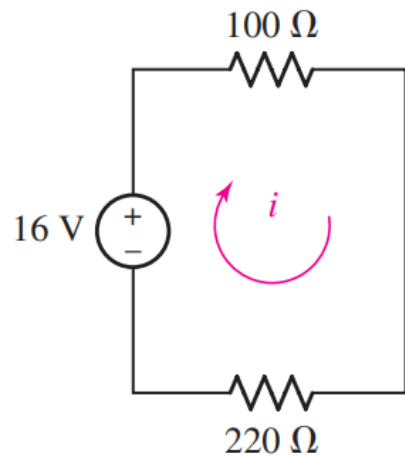
Ground potential.

FIG. 5.48

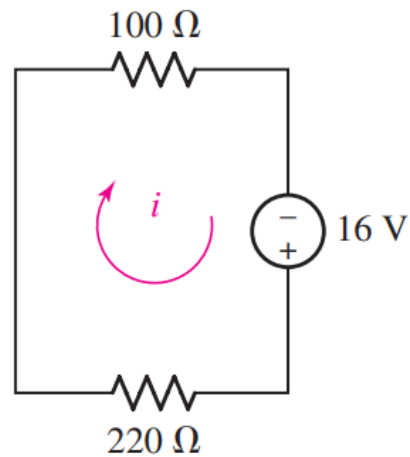
Three ways to sketch the same series dc circuit.



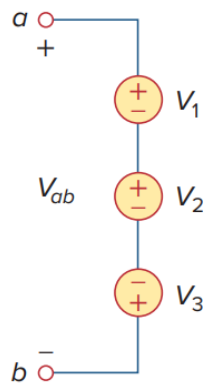
(a)



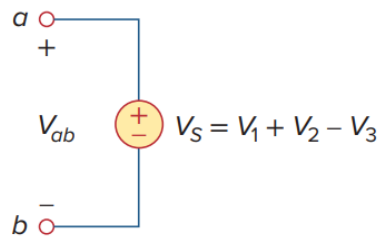
(b)



(c)



(a)



(b)

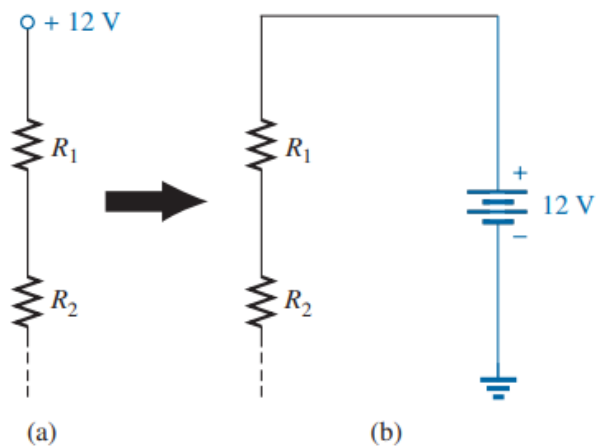


FIG. 5.49

Replacing the special notation for a dc voltage source with the standard symbol.

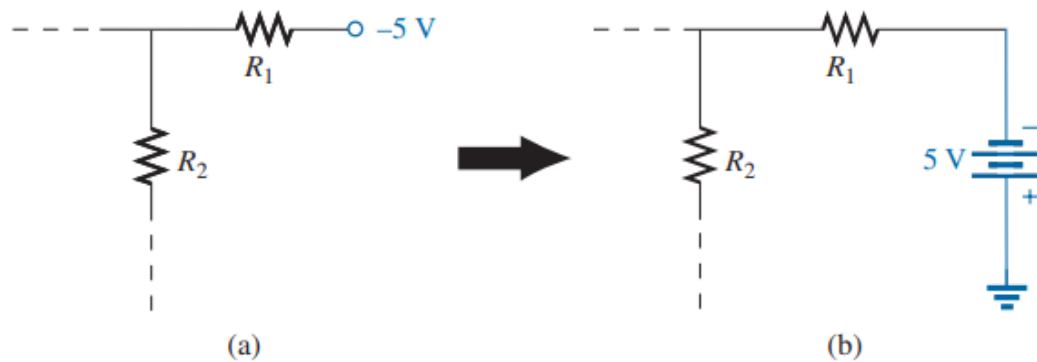
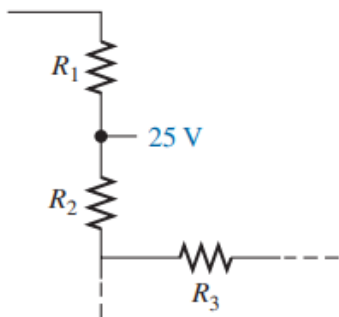


FIG. 5.50

Replacing the notation for a negative dc supply with the standard notation.



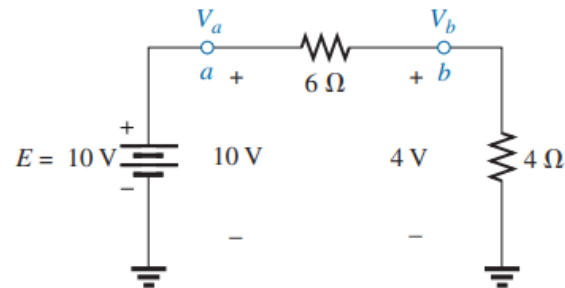


FIG. 5.53

Defining the use of single-subscript notation for voltage levels.

The single-subscript notation V_a specifies the voltage at point a with respect to ground (zero volts). If the voltage is less than zero volts, a negative sign must be associated with the magnitude of V_a .

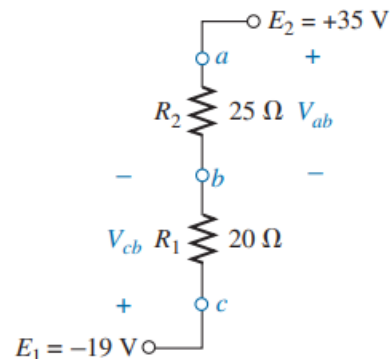


FIG. 5.61

Example 5.26.

Calculate the equivalent resistance R_{ab} in the circuit in Fig. 2.37.

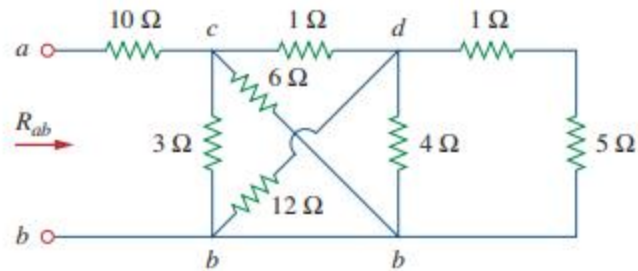


Figure 2.37
For Example 2.10.

Practice Problem 2.10

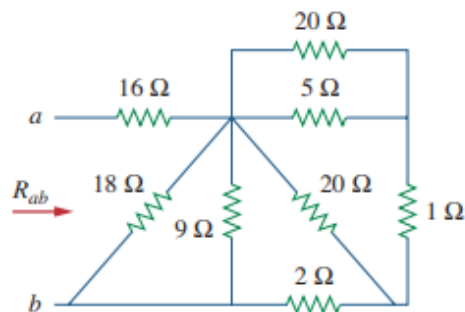


Figure 2.39
For Practice Prob. 2.10.

